

intentionally not on an NSBB). Lack of documentation, awaiting EGD or repeat Fibroscan, and missed or no scheduled follow-up accounted for cases of non-adherence to guidelines.

Conclusion: Adherence to NSBB initiation can be improved at our institution, with many eligible patients not prescribed NSBB without documented rationale. Outdated reliance on further invasive testing and missed or no scheduled follow-up contribute to non-adherence. Improved provider education and systematic prompts may enhance guideline-based NSBB prescribing and preventive care in portal hypertension.

S2879

Increase in Mortality and Allograft Rejection Post-Liver Transplant in a Cohort Between 50-90 Years Old

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Introduction: The proportion of the global population between 2015-2020 older than 60 years is predicted to nearly double from 12% to 22%. With an aging population, there has been a concomitant rise in the age of liver transplant (LT) recipients. Recipients aged 65 and older, are the fastest-growing cohort among new liver transplant registrants, surpassing the 35-49 age group over the past decade. There is an increasing need to understand post-operative outcomes in older patients who are at increased risk for frailty and with more comorbidities than their younger counterparts. This study aims to report clinical outcomes in a cohort of LT recipients aged 50 years and older.

Methods: This was a case series including 603 liver transplant recipients aged 50 years and older between 2016 and 2023. Non-identifiable information within the electronic health records (EHR) at academic hospital center was used. International Review Board approval was not required. ICD-9 Codes were utilized for data gathering from the database. EHR were mapped to the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM) version 5.3 such that standardized vocabularies like *International Classification for Diseases* (ICD) could be used to search for keywords. Comorbidity scores were calculated using the Charlson Comorbidity score.

Results: The study included 211 (35%) women with a mean age of 61.8 years. 43.3% of the cohort were White and 41.1% were Hispanic. The most common etiology of liver disease was alcohol (64.18%). The mean Charlson comorbidity index was 9.1, indicating severe comorbidity. 6.3% died within one year post LT increasing to 8.5% 2 years post LT. 22.2% of patients had LT rejection within one year post LT which increased to 24.7% 2 years post LT.

Conclusion: In this study of adults >50 years of age, there was a significant increase in mortality and rejection over time. These findings can be explained by graft rejection, infection and LT related complications over time. Although future prospective studies are needed to elucidate the underlying mechanisms, this study underscores the importance of studying LT outcomes in an older cohort of patients.

S2880

Racial Disparities in Liver Transplantation Among Patients With Primary Sclerosing Cholangitis

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Introduction: Primary sclerosing cholangitis (PSC) is a progressive hepatobiliary disease without effective medical therapies other than liver transplantation (LT). However, disparities in LT access and outcomes for minority patients with PSC remain poorly defined.

Methods: We conducted a retrospective cohort analysis using the United Network for Organ Sharing database from 2010-2024. PSC patients listed for LT were stratified by race (non-White vs White). Multivariable competing risks regression was used to investigate the primary outcome of time to waitlist dropout (due to being too sick) or mortality, accounting for LT as a competing risk. Secondary outcomes were time to post-LT graft failure, living donor LT, and listing for cholangiocarcinoma.

Results: Among 7,636 PSC patients evaluated, 5733 were White and 1903 were non-White; most non-White patients were Black (62%). Over time, the proportion of non-White patients with PSC listed for LT has risen. Non-White patients had higher MELD scores at listing (20 vs 17, $P < 0.001$) and were less likely to have concurrent inflammatory bowel disease (61% vs 73%, $P < 0.001$). Only 28 [1.5%] non-White patients with cholangiocarcinoma were listed for LT compared to 253 [4.4%] White patients ($P < 0.001$). Donation after circulatory death graft utilization was similar between the 2 groups; however, non-White patients were half as likely to undergo living donor LT (6% vs 12%, RR 0.5, $P < 0.001$). There was no difference in risk of waitlist dropout/mortality between White and non-White patients on competing risks regression (aHR 1.1, 95% CI 0.8-1.5, $P = 0.7$), though measures of socioeconomic status like education level (HR 0.8, 95% CI 0.7-0.9, $P < 0.001$) and public insurance (HR 1.4, 95% CI 1.1-1.7, $P < 0.001$) were associated with waitlist dropout/mortality. Race, education status, and insurance type were not associated with time to graft failure on Cox regression.

Conclusion: Racial disparities persist in access to liver transplantation for PSC, as non-White patients are less likely to receive living donor LT and less likely to be listed with concurrent cholangiocarcinoma. Despite these disparities, clinical outcomes were largely similar, suggesting resilience in care once access is achieved, though measures of socioeconomic status remained associated with waitlist outcomes. Interventions are needed to address barriers to early diagnosis, subspecialty referral, and living donor education in minority communities.

S2881

External Validation of a Fast-Track Extubation Score and Predictors of Extubation Time Following Liver Transplantation

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Introduction: Early extubation after liver transplantation (OLT) improves outcomes, but optimal timing remains unclear. The El Moheb Fast-Track Score (FTScore) was proposed to predict early extubation post-OLT. This study aimed to externally validate this score and to identify factors, including frailty, associated with time to extubation.

Methods: Adult OLT recipients at our institution from December 2019 to November 2024 were prospectively enrolled ($n = 288$). For FTScore validation, scores were calculated (MELD score < 29 [1 point], packed red blood cells [pRBC] < 4 units [2 points], fresh frozen plasma [FFP] < 4 units [1 point]; max 4 points), and the association with extubation within 4 hours and overall time to extubation was assessed. Preoperative data including Liver Frailty Index (LFI), Karnofsky Performance Scale (KPS), eGFR, and intraoperative factors such as blood product utilization were collected. Statistical analyses included Student's t-test, bivariate correlation, Kaplan-Meier analysis, and Cox proportional hazard modeling to identify predictors of time to extubation.

Results: Eighteen patients (6%) were extubated within 4 hours. They required less intraoperative pRBC (5.17 ± 3.43 vs 10.93 ± 9.02 , $P < 0.001$), FFP (2.94 ± 3.12 vs 8.71 ± 8.32 , $P < 0.001$), and total blood products (11.39 ± 9.24 vs 26.37 ± 21.24 , $P < 0.001$). However, the FTScore was not associated with extubation within 4 hours nor with overall time to extubation. Shorter time to extubation correlated with higher KPS ($r = -0.286$, $P < 0.001$), lower LFI ($r = 0.184$, $P = 0.006$), higher pre-transplant eGFR ($r = -0.166$, $P = 0.006$), male sex (16.8 ± 18.2 hours vs 23.1 ± 26.9 hours for women, $P = 0.034$), and lower total blood products ($r = 0.262$, $P < 0.001$). MELD 3.0 score, pre-transplant length of stay, race, and malnutrition were not significantly correlated with extubation time. Adjusting for sex, KPS, LFI, and eGFR, only LFI was an independent predictor of time to extubation (HR = 0.799 per unit LFI, 95% CI [0.685-0.931], $P = 0.004$). There was a significant difference in time to extubation across LFI categories (robust, pre-frail, frail; Log Rank $P = 0.011$).

Conclusion: The FTScore was not validated in our OLT cohort. However, our analysis identified LFI as a significant independent predictor of extubation time, with frailty being strongly associated with prolonged intubation. While scoring systems like the FTScore have benefits, LFI appears to be a more robust indicator for predicting post-OLT recovery milestones.

S2882

Revealing New Trends in the Global Burden of Liver Cancer Related to Hepatitis C Virus

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Introduction: Hepatocellular carcinoma (HCC) remains a leading cause of global cancer mortality, with a continually increasing incidence and poor 5-year survival. Hepatitis C virus (HCV) is a major risk factor for HCC, and while direct-acting antivirals (DAAs) have significantly reduced HCV-related HCC (HCV-HCC) risk, its global burden remains high. Given recent World Health Organization (WHO) initiatives to aimed at reducing hepatitis-related mortality, we evaluated trends in HCV-HCC from 1990 to 2021, stratified by gender, WHO region, and sociodemographic index (SDI), using data from the Global Burden of Disease 2021 study.

Methods: We extracted data on deaths, disability-adjusted life years (DALYs), and age-standardized incidence (ASI) attributable to HCV. Trends were analyzed using Joinpoint regression to calculate annual percent change (APC) and average annual percent change (AAPC), with significance set at $P < 0.05$.

Results: Globally, HCV-HCC deaths increased from 1990 to 2021 (woman AAPC 2.71 with 95% CI: 2.67-2.75 and man AAPC 2.65 with 95% CI: 2.60-2.70). Global DALYs also increased (woman AAPC 2.28 with 95% CI: 2.24-2.32 and man AAPC 2.16 with 95% CI: 2.10-2.21). In contrast, ASI declined (woman AAPC -0.51 with 95% CI: -0.53 to -0.51 and man AAPC -0.57 with 95% CI -0.61 to -0.53). In terms of WHO regions, DALYs rose across all WHO regions, with the Americas and Southeast Asia showing the highest increases. Deaths followed similar patterns, especially among men in the Americas and Eastern Mediterranean. ASI rates decreased in Africa, Western Pacific, and Eastern Mediterranean but rose in the Americas, Europe, and Southeast Asia. By SDI, middle and middle-low regions had the steepest increases in DALYs and deaths. High SDI regions saw the sharpest ASI declines after 2016, coinciding with WHO strategy implementation. ASI rates for women rose in high SDI settings and declined less than in men. Notably, ASI began to rise again after 2019 for both sexes.

Conclusion: Despite treatment advances, absolute HCV-HCC burden has grown unevenly. The increase in HCV-HCC deaths and DALYs with overall decreased ASI could indicate a rising disease burden due to population aging and growth. Regions with better resources saw ASI reductions, while others lagged. Gender and SDI disparities highlight gaps in access to care, screening, and treatment. Tailored interventions, improved equity, and sustained global efforts are essential to achieve hepatitis elimination goals.

S2883

Cigarette Smoking May Increase the Risk for Cirrhosis, Especially Among Women: Evidence From NHANES 2021-2023

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Introduction: Cigarette smoking significantly contributes to the global disease burden. While its impact on cardiovascular and pulmonary disease is well-established, the relationship with liver disease remains under-investigated. This study aims to evaluate the association between cigarette